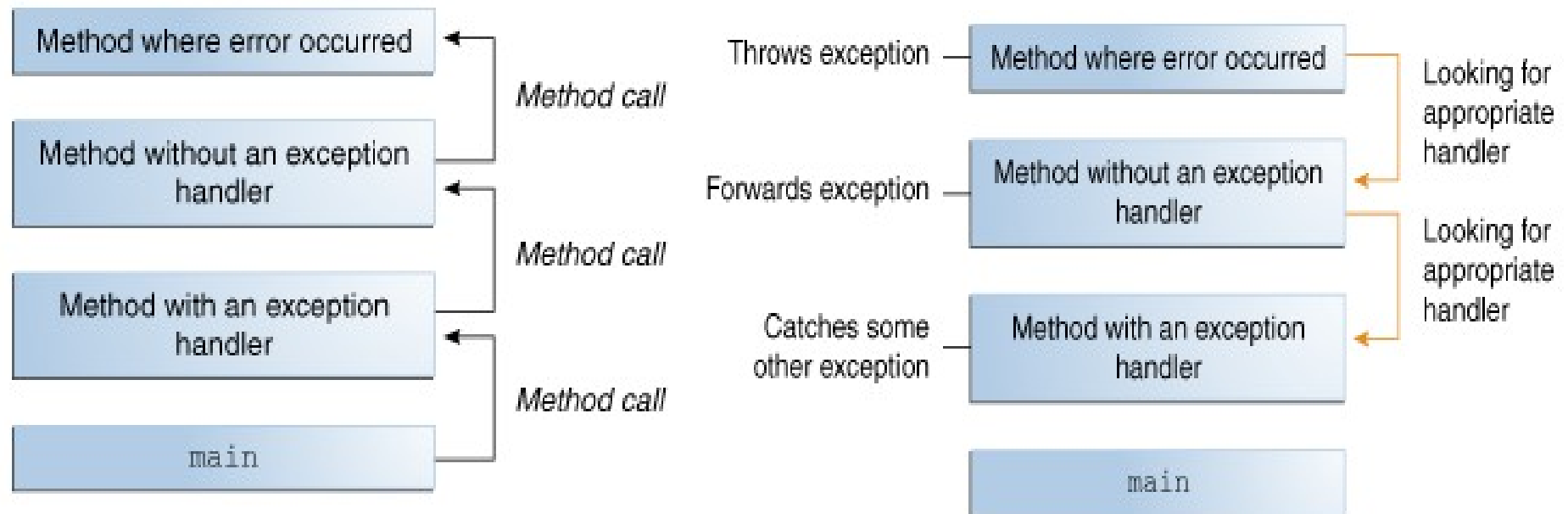


Exception Handling

Exception Handling

- Systematic approach to handle exceptions and errors in the program
 - Alternative to error codes
 - Allows better readability and maintainability of code



- Basic constructs
 - try, catch, finally, throw, throws

Need for Exception Handling

```
readFile {  
    open the file;  
    determine its size;  
    allocate that much memory;  
    read the file into memory;  
    close the file;  
}
```

Pseudocode for a file read operation

What happens if the file can't be opened?
What happens if the length of the file can't be determined?
What happens if enough memory can't be allocated?
What happens if the read fails?
What happens if the file can't be closed?

Possible issues

Example

```
errorCodeType readFile {
    initialize errorCode = 0;

    open the file;
    if (theFileIsOpen) {
        determine the length of the file;
        if (gotTheFileLength) {
            allocate that much memory;
            if (gotEnoughMemory) {
                read the file into memory;
                if (readFailed) {
                    errorCode = -1;
                }
            } else {
                errorCode = -2;
            }
        } else {
            errorCode = -3;
        }
        close the file;
        if (theFileDintClose && errorCode == 0) {
            errorCode = -4;
        } else {
            errorCode = errorCode and -4;
        }
    } else {
        errorCode = -5;
    }
    return errorCode;
}
```

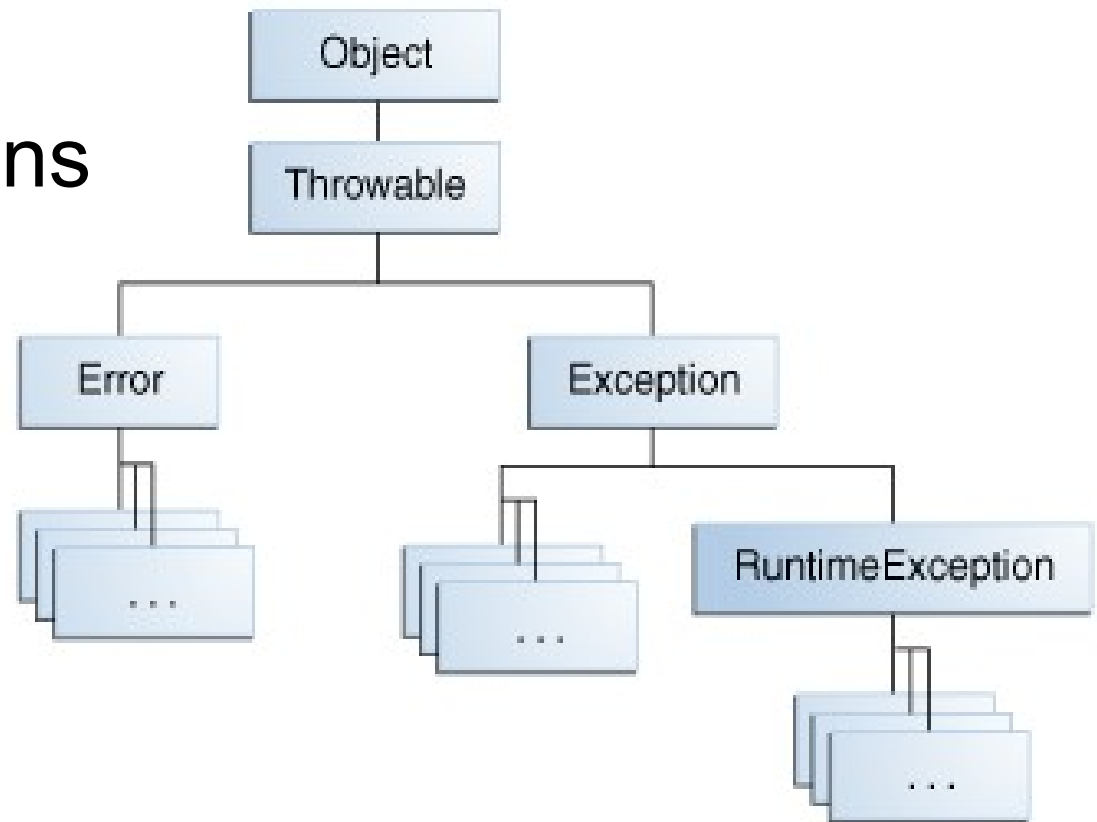
Error codes

```
readFile {
    try {
        open the file;
        determine its size;
        allocate that much memory;
        read the file into memory;
        close the file;
    } catch (fileOpenFailed) {
        doSomething;
    } catch (sizeDeterminationFailed) {
        doSomething;
    } catch (memoryAllocationFailed) {
        doSomething;
    } catch (readFailed) {
        doSomething;
    } catch (fileCloseFailed) {
        doSomething;
    }
}
```

Exception Handlers

Exception Types (in Java)

- Checked Exceptions
- Unchecked Exceptions
 - Runtime Exceptions
 - Error



Guidelines

- Use appropriate handlers
 - Propagate exceptions to appropriate handlers where necessary
 - Don't swallow exceptions
- Log at the minimum to investigate what went wrong